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1.

Let $\mathcal{R}$ be a binary relation on a set A . Which one of the following is true about the relation $\mathcal{R}$?

1 / 1 point

☐

$\mathcal{R}$ is symmetric if and only if $\forall x, y, z \in A$, if $x\mathcal{R}y$ and $y\mathcal{R}z$ then $x\mathcal{R}z$

☐

$\mathcal{R}$ is symmetric if and only if $\forall x, y \in A$, if $x\mathcal{R}y$ and $y\mathcal{R}x$ then $x = y$

☒

$\mathcal{R}$ is symmetric if and only if $\forall x, y \in A$, if $x\mathcal{R}y$ then $y\mathcal{R}x$

☐

$\mathcal{R}$ is symmetric if and only if $\forall x \in A, x\mathcal{R}x$

☒

Nice work

Correct. This is the definition of a symmetric relation.
2.

Let $\mathcal{R}$, a binary relation on $\mathbb{Z}$ (a set of all integers), be defined as $a\mathcal{R}b$ if and only if $(a - b) \bmod 2 = 0, \forall a, b \in \mathbb{Z}$. Which one of the following statements is true about the relation $\mathcal{R}$?

1 / 1 point

☐

$\mathcal{R}$ is reflexive and anti-symmetric

☐

$\mathcal{R}$ is symmetric but not reflexive

☒

$\mathcal{R}$ is reflexive and symmetric

☐

$\mathcal{R}$ is reflexive but not symmetric

☒

Nice work

Correct. Since $(x - x) \bmod 2 = 0$ and $(x - y) \bmod 2 = (y - x) \bmod 2$, the relation is both reflexive and symmetric.
3.

Let $M_{\mathcal{R}} = [m_{ij}]$ be a representing a symmetric binary relation, $\mathcal{R}$, on a set A . Which one of the following is true?

1 / 1 point

☐

If $i \neq j$ and $m_{ij} \neq 0$ then $m_{ji} = 0$

☐

The leading diagonal of $M_{\mathcal{R}}$ contains 0s only

☐

The leading diagonal of $M_{\mathcal{R}}$ contains 1s only

☒

If $i \neq j$ and $m_{ij} \neq 0$ then $m_{ji} \neq 0$

☒

Nice work

Correct. This indicates that the matrix is symmetric. Hence, the relation is symmetric.
4.

Let $G_{\mathcal{R}}$ be a digraph representing a symmetric binary relation, $\mathcal{R}$, on a set A . Which one of the following is true?

1 / 1 point

☐

Every vertex in $G_{\mathcal{R}}$ contains a loop

☐

Given any two different vertices v_i and v_j in $G_{\mathcal{R}}$, if there is an arrow from v_i to v_j , then there is no arrow from v_j to v_i

☒

Given any two different vertices v_i and v_j in $G_{\mathcal{R}}$, if there is an arrow from v_i to v_j , then there must be an arrow from v_j to v_i

☐

None of the the vertices in $G_{\mathcal{R}}$ contains a loop

☒

Nice work

Correct. The relation is symmetric, hence every pair of different vertices in its corresponding digraph are either linked both ways or not linked at all.